

3D Gaussian Splatting vs. Classical UAV Photogrammetry: A Systematic Comparative Analysis for Drone-Based Reconstruction

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Executive Summary

The central finding of this review is that 3DGS and classical photogrammetry optimize for fundamentally different objectives, and no single pipeline dominates across all evaluation dimensions. The literature through mid-2026 consistently demonstrates that classical SfM/MVS achieves superior metric geometric accuracy (1.43 mm RMSE vs. 4.74–7.28 mm for 3DGS variants in controlled conditions [50]), while 3DGS delivers substantially higher visual quality (PSNR, SSIM, LPIPS) and real-time rendering at 100+ FPS [10] [43]. The critically counterintuitive GauU-Scene V2 finding—that NeRFacto achieved the best geometric accuracy (0.093 m Chamfer distance) despite the worst rendering metrics (PSNR 17.04 dB), while vanilla 3DGS achieved the best rendering metrics (PSNR 24.42 dB) but inferior geometric accuracy (0.133 m)—should anchor the framing of any comparative thesis [101].

Key quantitative thresholds:

- Classical SfM-MVS geometric RMSE: **1.43 mm** (no occlusion) / **3.23 mm** (vegetation) [50]
 - Best 3DGS geometric RMSE: **4.74 mm** (3DGS-Basic, no occlusion) / **7.49 mm** (vegetation) [50]
 - GeoRefGS georeferencing accuracy: **0.054 m** planar / **0.053 m** vertical—an improvement of 83% over SfM-BBA baseline of 0.32 m [63]
 - Skyfall-GS is **satellite-only** (WorldView-3/GoogleEarth data)—it has not been tested on UAV/drone footage and is architecturally incompatible with standard oblique aerial workflows without significant adaptation [5] [7]
 - The confirmed research gap: **no published study** has combined a physical GCP testfield measured by total station/RTK GNSS with 3DGS reconstruction and evaluated metric accuracy against surveying-grade ground truth
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1. 3DGS Fundamentals and Variant Landscape — Theoretical Foundations for a Geodetic Comparison

3DGS optimizes for visually faithful rendering; geometric accuracy requires explicit additional constraints that most implementations omit.

1.1 Foundational Architecture

Kerbl, Kopanas, Leimkühler, and Drettakis (2023) introduced "3D Gaussian Splatting for Real-Time Radiance Field Rendering" in *ACM Transactions on Graphics*, Volume 42(4), awarded Best Paper at SIGGRAPH 2023 [10] [11]. The paper is affiliated with Inria, Université Côte d'Azur, and Max-Planck-Institut für Informatik, with the arXiv identifier 2308.04079 [10]. As of the SciSpace aggregator's count it has accumulated approximately 1,414 citations [54], though this figure is certainly higher given its rapid adoption.

Three architectural innovations define the method: (1) anisotropic 3D Gaussian ellipsoid scene representation initialized from SfM sparse points, enabling continuous volumetric radiance field properties without computation in empty space; (2) interleaved optimization and adaptive density control with optimized anisotropic covariance for accurate scene representation; and (3) a tile-based differentiable rasterizer supporting anisotropic splatting that achieves ≥ 30 FPS at 1080p resolution [9] [10]. The rendering paradigm departs entirely from NeRF's ray-marching through an MLP and instead employs GPU rasterization over explicit geometric primitives, enabling real-time performance exceeding 100 FPS on consumer hardware [43] [45].

The fundamental geodetic limitation is implicit in this architecture: 3DGS optimizes photometric loss—the difference between rendered and observed pixel values—not geometric conformity to a physical surface. Gaussians can deform, overlap, and aggregate in configurations that appear visually correct but deviate metrically from the underlying geometry, particularly in textureless regions, under vegetation, and at object boundaries.

1.2 Architecturally Relevant Variants

The following variants are directly relevant to the thesis evaluation framework and each addresses a specific limitation of the base implementation:

Mip-Splatting (CVPR 2024 Oral, Best Student Paper; Yu et al., arXiv:2311.16493) [13] [15] addresses aliasing when camera distance or focal length changes—a critical issue for UAV imagery spanning multiple altitudes. Two components: a 3D Smoothing Filter constraining Gaussian primitive sizes based on maximal sampling frequency from input views, and a 2D Mip

Filter replacing 2D dilation with a box filter approximating physical imaging. On the Blender dataset (multi-scale), Mip-Splatting achieves average PSNR 34.56 dB versus vanilla 3DGS's 28.79 dB at full resolution [15].

2D Gaussian Splatting (2DGS) (SIGGRAPH 2024; Huang et al., arXiv:2403.17888) [14] [53] collapses the 3D volume into 2D oriented planar Gaussian disks, providing view-consistent geometry and intrinsic surface modeling via ray-splat intersection and rasterization. This architectural choice is the most geodetically relevant variant: in a controlled transparent object experiment, 2DGS achieved RMSE 0.00121 m (1.21 mm) versus 3DGS+SuGaR's 0.00125 m and Agisoft Metashape's 0.00161 m—2DGS was geometrically superior despite inferior rendering metrics (PSNR 34.91 dB vs. 3DGS's 36.06 dB) [74]. The lower standard deviation of 2DGS (0.0005 m vs. 0.0011 m for 3DGS+SuGaR) indicates more spatially consistent accuracy [74].

Scaffold-GS (CVPR 2024 Highlight; Lu et al., arXiv:2312.00109) [105] [117] uses anchor points from a sparse voxel grid with neural Gaussians spawned via learnable offsets, dynamically predicting attributes (opacity, color, scale, rotation) from viewing direction. The primary advantage is storage efficiency: 156 MB vs. 693 MB on Mip-NeRF360 (4.4× reduction), 87 MB vs. 411 MB on Tanks&Temples (4.7× reduction) [117]. Rendering speed is maintained at approximately 100 FPS. On Mip-NeRF360, Scaffold-GS achieves PSNR 28.84 vs. 3DGS's 28.69—comparable rendering quality with dramatically reduced memory footprint [117].

SuGaR (CVPR 2024; Guédon & Lepetit, arXiv:2311.12775) [103] [104] introduces surface-alignment regularization that encourages Gaussians to align with scene surfaces and be well-distributed above them, followed by Poisson reconstruction (depth 10) for clean mesh extraction in 5–10 minutes per scene. Total optimization requires 15–45 minutes including refinement. On Mip-NeRF360, the refined variant R-SuGaR-15K achieves indoor PSNR 29.43, SSIM 0.910, LPIPS 0.216; outdoor PSNR 24.40, SSIM 0.699, LPIPS 0.301 [103]. SuGaR's primary relevance for geodetic work is enabling mesh output from 3DGS scenes for CAD/GIS compatibility—though mesh extraction introduces additional approximation errors relative to the splat representation.

GaussianPro (ICML 2024; Cheng et al., arXiv:2402.14650) [55] [56] [110] introduces a progressive propagation strategy using MVS patch-matching to generate new Gaussians with precise positions and orientations, specifically addressing 3DGS's dependence on SfM initialization in textureless regions. On the Waymo dataset (scene 1002751), GaussianPro achieves 35.97 PSNR, 0.959 SSIM, 0.207 LPIPS versus vanilla 3DGS's 35.22 PSNR—a 1.15 dB improvement [57]. Training increases from 40 to 56 minutes and rendering slightly decreases from 119 to 108 FPS [111].

3DGS-DR (SIGGRAPH 2024; Ye et al., arXiv:2404.18454) [115] [118] implements deferred shading for specular reflections in a two-pass architecture: first pass renders Gaussian properties (base color, normal, reflection strength), second pass performs pixel-level shading with environment map queries. On the Shiny Blender dataset, the deferred variant achieves average PSNR 33.66, SSIM 0.979, LPIPS 0.098—substantially outperforming the forward variant (PSNR 27.64). Training requires 16 minutes versus Ref-NeRF's 19 hours, and rendering achieves 251 FPS versus GaussianShader's 51 FPS [118].

gsplat (JMLR 2024) [16] is an open-source PyTorch+CUDA library providing 10% less training time and 4× less memory than the original Kerbl et al. reference implementation, with support for multiple densification strategies (ADC, gradient-based, MCMC) and integrated Mip-Splatting anti-aliasing. This is the recommended implementation for a THWS student with access to standard GPU hardware.

1.3 Surveys and Standardization Context (2024–2026)

Five surveys provide the literature backbone: Chen & Wang (arXiv:2401.03890) [20] [21], Fei et al. (IEEE TVCG, DOI: 10.1109/TVCG.2024.3397828) [17], arXiv:2402.07181 with 6-aspect taxonomy [22], He et al. (arXiv:2508.09977, segmentation/editing/generation applications) [18], and MDPI Sensors 25(12):3626 (2025) [23]. The standardization context in 2026 is defined by: the KHR_gaussian_splatting Release Candidate for glTF 2.0 (announced February 2026, ratification expected Q2 2026) developed by Khronos Group, OGC, Niantic Spatial, Cesium, and Esri [66] [69]; the SPZ compression format (MIT license, Niantic, up to 90% vs. PLY) [67]; and OpenUSD v26.03's UsdVolParticleField3DGaussianSplat schema from AOUSD (Pixar, Apple, NVIDIA, Adobe) [19].

2. Skyfall-GS: Critical Scoping Assessment — A Satellite-Only Architecture Not Transferable to UAV Without Adaptation

Skyfall-GS is architecturally optimized for nadir satellite imagery and has not been evaluated on oblique UAV drone footage. This is the most important practical finding for thesis scoping.

2.1 Full Citation and Architecture

Lee, J.-Y., Liu, Y.-R., Tsai, S.-R., Chang, W.-C., Wu, C.-H., Chan, J., Zhao, Z., Lin, C.H., & Liu, Y.-L. (2025). "Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery." arXiv:2510.15869, submitted October 17, 2025. Currently under conference review (OpenReview). *Status: arXiv preprint, not yet peer-reviewed*. Institutional affiliations: National Yang Ming Chiao Tung University (lead), UIUC, University of Zaragoza, UC Merced. GitHub: <https://github.com/jayin92/skyfall-gs> (843 stars) [1] [5] [49].

The two-stage pipeline architecture consists of:

Stage 1 (Reconstruction, ~1 hour on RTX A6000): Enhanced 3DGS with three satellite-specific components: (a) pseudo-camera depth supervision using MoGe monocular depth estimation from virtual cameras at reduced elevations (80° down to 45°), sampled every 10 iterations across 24 views; (b) WildGaussians-adapted appearance MLP with 2 hidden layers (128 neurons, ReLU), per-image embeddings (32 dimensions), and per-Gaussian embeddings (24 dimensions); (c) entropy-based opacity regularization ($\lambda_{op} = 10$). Modified densification: gradient threshold lowered from 0.002 to 0.001; scaling learning rate reduced from 0.005 to 0.001; Gaussian pruning threshold at maximum covariance > 20 [5].

Stage 2 (Curriculum refinement, ~6 hours on RTX A6000): 5 episodes $\times$ 10,000 iterations. Uses FLUX.1 [dev] diffusion model with FlowEdit prompt-to-prompt editing, transforming satellite artifacts to "clear satellite image with sharp buildings, smooth edges, and natural lighting." Multi-sample strategy: $N_s = 2$ independent refined samples per view. Curriculum view progression from 85° elevation descending to 45° [5].

2.2 Performance Metrics

On the DFC2019 dataset (Jacksonville, FL; WorldView-3 imagery): Skyfall-GS achieves FIDCLIP 27.35, CMMD 2.086, PSNR 12.38, SSIM 0.321, LPIPS 0.791 versus Sat-NeRF (FIDCLIP 88.36), EOGS (87.74), Mip-Splatting (87.19), and CoR-GS (89.03)—all substantially worse [5]. On the GoogleEarth dataset (NYC): FIDCLIP 9.91, CMMD 2.009, PSNR 14.28, SSIM 0.298, LPIPS 0.394 versus GaussianCity (28.73) and CoR-GS (27.32) [5]. Lower FIDCLIP and CMMD indicate better perceptual quality using distribution-based CLIP-backbone metrics. Rendering: 11 FPS on NVIDIA T4; 40 FPS on MacBook Air M2 [4]. Human evaluation with 89 participants: ~97% win rate on DFC2019, ~90% on GoogleEarth [4] [5].

The training data consists of Jacksonville scenes (JAX_004: 9 images, JAX_068: 17 images, JAX_214: 21 images, JAX_260: 15 images) and NYC scenes (4 areas with 60 training images each) [2].

2.3 Fundamental Thesis-Scoping Limitations

The following limitations, several explicitly stated in the paper and one confirmed by community discussion, define why Skyfall-GS cannot be directly used as a UAV pipeline:

- Satellite-only validation:** The pipeline is designed for near-nadir WorldView-3 and GoogleEarth imagery with extremely limited parallax. No evaluation on UAV or oblique aerial imagery exists in the paper [5].
- No geodetic accuracy evaluation:** Skyfall-GS uses distribution-based metrics (FIDCLIP, CMMD) and user studies. No GCP validation, no RMSE against surveying instruments, no Chamfer distance against LiDAR [5]. This is a fundamental incompatibility with geodetic benchmarking.
- Generative hallucination risk:** The FLUX.1 [dev] diffusion refinement stage introduces the risk of synthesizing details inconsistent with actual ground-level geometry [7]. Community feedback confirms: "when you bring the viewpoint down to near street level and look horizontally, it's worse than traditional photogrammetry methods" [7].
- Over-smoothed street-level textures** at near-horizontal viewing angles, acknowledged in the paper [5].
- Diffusion-dependent processing time:** Total ~7 hours on an RTX A6000, primarily driven by the generative refinement stage—not directly comparable to the photogrammetric SfM pipeline [5].

For the thesis, the student should clarify this scoping issue in the PA phase. More appropriate drone-specific 3DGS pipelines are **DroneSplat** (CVPR 2025, arXiv:2503.16964), the standard INRIA 3DGS implementation, DJI Terra V5.0 for commercial benchmarking, or the large-scale UAV pipeline from arXiv:2602.20342 [24] [25] [26] [40].

3. Classical UAV Photogrammetry: Accuracy Benchmarks and Software State of the Art

Classical SfM/MVS remains the geometric accuracy standard for UAV mapping, with checkpoint RMSEs of 1.8–2.7 cm achievable at typical UAV flight parameters using commercial software.

3.1 Software Benchmark Comparison (2024)

A processing benchmark using an identical 420-image dataset (48 acres, 75% front overlap, 65% side overlap, DJI Mavic 3 Enterprise at 80 m AGL) provides direct comparability [46]:

Software	Checkpoint RMSE	Point Density	Total Processing	License (2026)
Pix4D Mapper	1.8 cm	412 pts/m ²	2h 47m	\$4,200/yr (cloud)
Agisoft Metashape Pro	1.9 cm	389 pts/m ²	2h 12m	\$3,499 perpetual
DroneDeploy	2.4 cm	298 pts/m ²	54 min (cloud)	\$4,000–\$12,000/yr
OpenDroneMap (WebODM)	2.7 cm	341 pts/m ²	3h 38m	Free (self-hosted)
RealityCapture	—	—	10–20× faster than Metashape (>1K img)	€250/3 months (PPI model)

Metashape Pro's perpetual license achieves break-even with Pix4D's subscription within 10 months [46]. DroneDeploy's cloud-first model eliminates workstation hardware investment and reduces labor monitoring overhead, achieving ~50% lower cost-per-deliverable (\$36.93 vs. \$72.77) when processing ≥15 projects/month [46]. COLMAP remains the scientific standard for maximum precision at zero licensing cost [47].

For 500-image building facade datasets, processing times on i9-12900K + RTX 4080: COLMAP + OpenMVS ~6 hours, OpenSfM ~4 hours, Agisoft Metashape ~3 hours [72]. The 2022 peer-reviewed comparison in *Drones* journal (Pell et al., Drones 2022; 6(1):24) directly compared Metashape, Pix4D, WebODM, and Correlator3D, confirming their broadly comparable outputs with software-specific workflow strengths [48].

3.2 GCP-Based Georeferencing Accuracy

The foundational GCP study (MDPI Remote Sensing 2018, 3,465 GCP combinations across 1,225 hectares [31]) establishes these relationships:

GCP Count	RMSE	Relation to GSD
10–20 GCPs	±31 cm	≈5× GSD
50–60 GCPs	±16 cm	≈3× GSD
90–100 GCPs (optimal)	±12 cm	≈2× GSD
≥3–5 GCPs per 100 photos	±GSD (planimetric)	≈1× GSD

Vertical errors are approximately **2.5× greater than horizontal errors** [31]. The critical methodological distinction: RMSE measured on control points (GCPs used in bundle adjustment) overestimates true accuracy by factors of 3–6× compared to independent check points [31]. With only 3–5 GCPs, control point residuals may show millimeter-level values while independent check points reveal actual RMSE of ±1–8 m [31].

RTK/PPK integration substantially changes the GCP calculus. Direct georeferencing with DJI Phantom 4 RTK achieves 1–3 cm horizontal, 4–6 cm vertical without GCPs [30]. On the Santa Rita Experimental Range deployment (38 m AGL, 2019): 2.2 cm horizontal RMSE and 3.4 cm vertical RMSE [58]. Combined RTK with nadir + oblique imagery (no GCPs): 1.26 cm XY RMSE and 5.58 cm Z RMSE—nadir-only RTK alone degrades to 22.97 cm vertical error [30].

Integrating oblique images (20° to nadir) as 10–12% of nadir images improves accuracy to 0.0141 m with precision of 0.03396 m [29]. Camera angle is more important than flying height for the spatial structure of errors; higher inclinations (20–40°) reduce both error magnitude and spatial autocorrelation (Moran's I) [32].

3.3 Video-Based SfM Workflows

Video-based photogrammetry reduces field capture time by 50–70% compared to still image acquisition for small-to-medium objects [38]. The 2024 Wuhan University VSLAM+SfM hybrid (DJI M300 RTK with DJI P1, 3840×2160 @ 60 fps) achieves 100–200% efficiency improvement versus COLMAP and 50–100% versus Theia, with centimeter-level accuracy under GCP constraints [41]. 8K video improves point cloud density and reconstruction accuracy up to 90% versus HD video from the same drone and altitude, though storage and compute requirements increase >20× [39].

For drone-class cameras (DJI Mini 2), image quality between raw video screenshots and dedicated still photography is minimal—photogrammetry software detects no significant difference [42].

4. 3DGS on UAV/Aerial Data: Specific Implementations and Performance

Drone-specific 3DGS implementations have emerged rapidly since 2024, with methods now addressing the core challenges of unbounded scenes, dynamic objects, limited parallax, and large-scale partitioning.

4.1 Key Drone-Specific Methods

DroneSplat (CVPR 2025; Tang et al., arXiv:2503.16964) [25] [106] directly addresses in-the-wild drone reconstruction with an Adaptive Local-Global Masking system for dynamic objects and Multi-View-Stereo predictions with voxel-guided optimization. The paper introduces a 24-sequence dataset captured with DJI Mavic Pro 3 at 1920×1080 and 3840×2160. On the "Low Dynamic" Cultural Center scene: PSNR 24.53, SSIM 0.785, LPIPS 0.149 versus GS-W (PSNR 23.29). On "High Dynamic" Simingshan: PSNR 24.17, SSIM 0.808, LPIPS 0.093 versus vanilla 3DGS (22.14). Under 6-view limited input, DroneSplat achieves PSNR 18.09–18.17, SSIM 0.496–0.534—the best performance compared to Scaffold-GS, FSGS, and InstantSplat [27] [106].

HUG: Hierarchical Urban Gaussian Splatting (ICCV 2025; Su et al., arXiv:2504.16606) [33] [34] implements octree-based Level-of-Detail with visibility-based partitioning (threshold $\tau_p = 800$ visible points per block), reducing preprocessing from ~2 hours to ~1 minute per scene on the MatrixCity dataset. HUG achieves PSNR 31.12 vs. CityGS baseline 27.23, SSIM 0.912 vs. 0.806, LPIPS 0.106 vs. 0.307 [33].

AGS: Aerial Gaussian Splatting (2024) [36] [37] was the first application of 3DGS to large-scale aerial MVS, evaluated on the WHU-OMVS dataset (10-cm GSD, 550 m altitude, 268 images at 3712×5504 px, 850×700 m² coverage) and the Tianjin dataset (20-cm GSD, 200 m altitude, 342 images at 3840×2560 px, 400×350 m² coverage, 80% flight-direction overlap). AGS surpassed GOF by 1.5% and COLMAP by 9.4% in highest-resolution PAG metrics [36] [37].

EA-3DGS (arXiv:2505.10787) [12] addresses initialization in weak-texture regions via adaptive tetrahedral mesh guidance. On the SCUT Campus Aerial dataset (595 images, DJI Matrice 300 RTK, ~100 m altitude, downsampled to 1600×841) it achieves PSNR 30.53, SSIM 0.902, LPIPS 0.130 on the Tower-Two scene versus Mega-NeRF's PSNR 20.93—a 9.6 dB improvement [12].

Large-scale Photorealistic UAV 3DGS (arXiv:2602.20342, 2026, Horizon Europe TRIFFID project) [26] [40] implements real-time near-live 3D model updates via RTMP streaming, semantic segmentation, COLMAP-style SfM, and the INRIA 3DGS implementation on AMD Ryzen 9 9900X3D + NVIDIA RTX 4070 (12 GB) + 64 GB RAM. Benchmark results: Mip-NeRF360: SSIM 0.815, PSNR 27.21, LPIPS 0.214 at 134 FPS; Tanks-and-Temples: SSIM 0.841, PSNR 23.14, LPIPS 0.183 at 154 FPS; Deep Blending: SSIM 0.903, PSNR 29.41, LPIPS 0.243 at 137 FPS. Reconstruction quality remains within 4–7% of high-fidelity offline references [40].

DJI Terra V5.0 (July 17, 2025) [24] marks the first integration of 3DGS into consumer drone processing software. Processing capacity: ~500 images/hour (versus ~200 images/hour in Terra 4.0); 1,500-image urban mapping projects complete in <3 hours versus 6–8 hours with mesh technology; datasets >10,000 images achieve 2× speedup. Memory: 300–400 images/GB RAM; 64 GB workstation handles 20,000–25,000 images; 128 GB handles 40,000. Optimized for Matrice 4E (RTK), Matrice 400 series, Zenmuse P1 (45 MP full-frame), and Zenmuse L2 (LiDAR) [24].

ULSR-GS (arXiv:2412.01402) [35] [116] handles scenes >1 km² via point-to-photo partitioning and multi-view geometric consistency constraints with adaptive multi-view densification. Evaluated on GauU-Scene (6 regions, >1 km² each, ~1,200 oblique images per region), UrbanScene3D (5,000 Penta-Cam images within 2.5 km), and custom urban datasets (~4 km², >8,000 campus images; ~2 km², ~10,000 oblique images). Building scene PSNR 23.49, SSIM 0.812, LPIPS 0.142 [116].

4.2 The GauU-Scene V2 Benchmark: Critical Geodetic Insight

The GauU-Scene V2 benchmark (arXiv:2404.04880) [101] is the most relevant large-scale UAV 3DGS dataset with LiDAR ground truth. Captured with DJI Matrix 300 drone + Zenmuse L1 LiDAR, the dataset covers 6 scenes across 6.668 km² total: Lower Campus (1.020 km², 670 images, 79.8M LiDAR points), Upper Campus (0.923 km², 715 images, 94.2M points), HAV (0.815 km², 424 images, 26.8M points), LFLS (1.467 km², 1,106 images, 98.5M points), SMBU (0.908 km², 563 images, 283.3M points), SZIIT (1.557 km², 1,215 images, 59.0M points). Total: 4,693 images, 627.5M LiDAR points, 92.1 GB raw data [101].

The paradigm-shifting finding:

Method	PSNR (dB)	SSIM	LPIPS	Chamfer Distance (m)	Training Time (GPU-min)
Vanilla 3DGS	24.42 (best)	0.752 (best)	0.300 (best)	0.133	63.7 (fastest)
NeRFacto	17.04 (worst)	0.422 (worst)	0.764 (worst)	0.093 (best)	1,735.3
InstantNGP	19.74	0.510	0.815	0.149	—
SuGaR	—	—	—	0.189 (worst)	—

This inverse relationship between rendering quality and geometric fidelity should be the conceptual anchor for any geodetically motivated comparison [101]. 3DGS optimizes appearance; without explicit geometric constraints (depth supervision, GCP-based loss, surface regularization), high PSNR does not imply accurate metric reconstruction.

5. Direct Comparative Studies: 3DGS vs. Classical Photogrammetry

5.1 ISPRS Annals 2025 — Most Geodetically Relevant Study

Petrovska, I. and Jutzi, B. (2025). "3D Gaussian Splatting Methods for Real-World Scenarios." *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-G-2025, pp. 641–648. DOI: 10.5194/isprs-annals-X-G-2025-641-2025. Conference: ISPRS Geospatial Week 2025, April 6–11, Dubai. Authors: Karlsruhe Institute of Technology, Institute for Photogrammetry and Remote Sensing [50] [112] [113].

This study provides the single most geodetically meaningful comparison in the available literature, evaluating four 3DGS implementations (3DGS-Basic, Splatfacto, 3DGS-MCMC, 3DGS-LumaAI) against MVS and NeRF on the STELLA dataset, with specific focus on vegetation occlusion as a critical real-world scenario for forestry and agricultural monitoring.

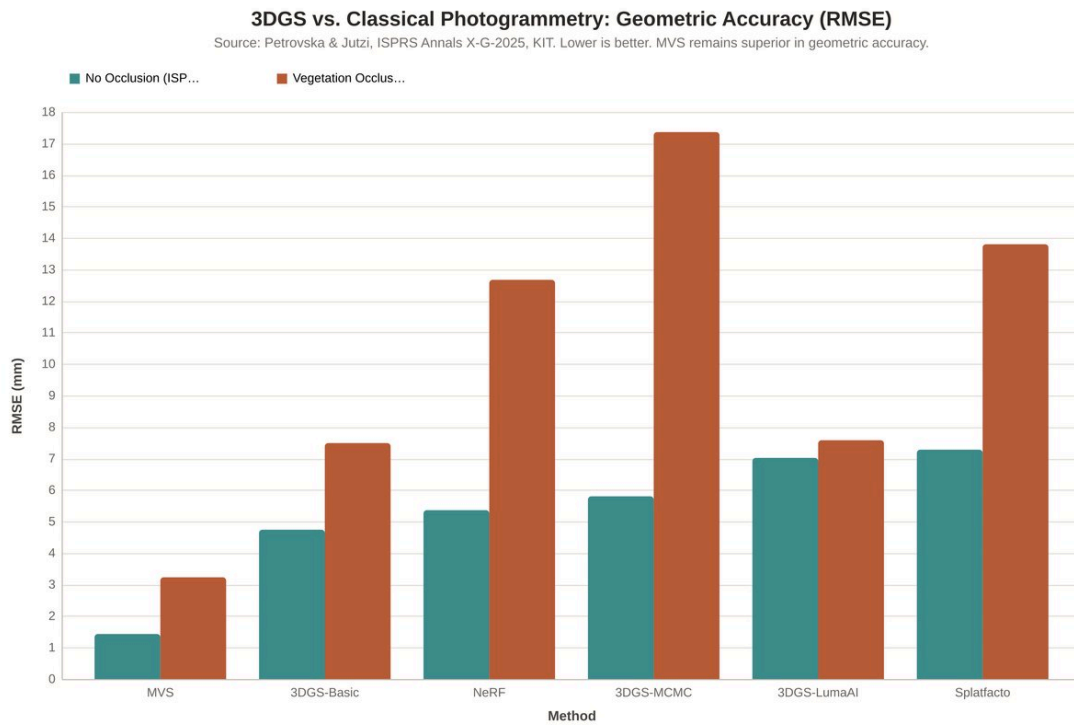


Figure 1: 3DGS vs. Classical Photogrammetry: Geometric Accuracy (RMSE): Source: Petrovska & Jutzi, ISPRS Annals X-G-2025, KIT. Lower is better. MVS remains superior in geometric accuracy.

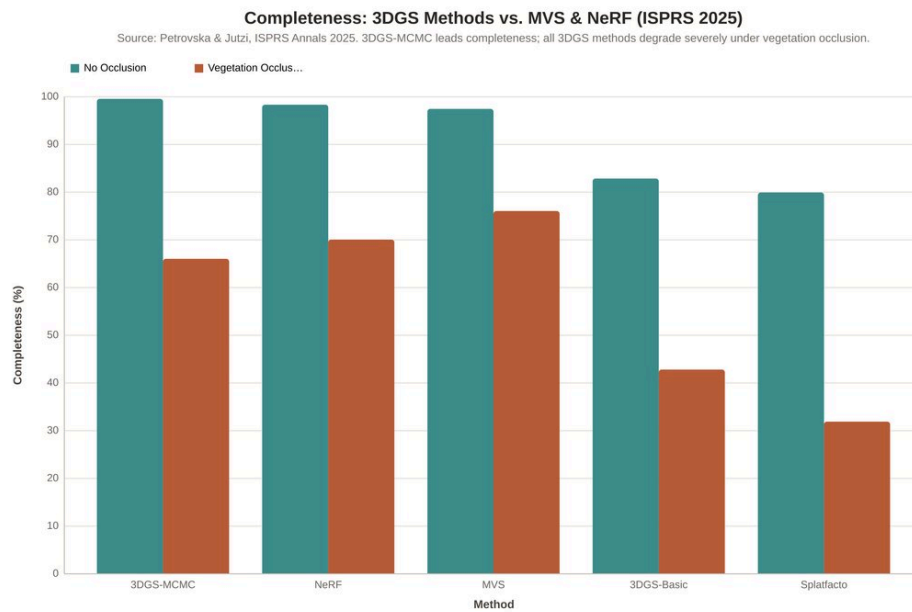


Figure 2: Completeness: 3DGS Methods vs. MVS & NeRF (ISPRS 2025): Source: Petrovska & Jutzi, ISPRS Annals 2025. 3DGS-MCMC leads completeness; all 3DGS methods degrade severely under vegetation occlusion.

Geometric Accuracy (RMSE in mm):

Method	No Occlusion	Vegetation Occlusion
MVS	1.43 mm	3.23 mm
3DGS-Basic	4.74 mm	7.49 mm
3DGS-MCMC	5.80 mm	17.36 mm (worst)
NeRF	5.36 mm	12.67 mm
3DGS-LumaAI	7.02 mm	7.58 mm
Splatfacto	7.28 mm (worst)	13.80 mm

Completeness (%):

Method	No Occlusion	Vegetation Occlusion
3DGS-MCMC	99.45%	65.93%
NeRF	98.23%	69.95%
MVS	97.35%	75.95%
3DGS-Basic	82.75%	42.71%
Splatfacto	79.83%	31.80% (worst)
3DGS-LumaAI	—	27.02% (worst)

Point cloud density: 3DGS-MCMC reconstructed 1,556,040 points versus 845,456 for MVS and 352,298 for 3DGS-Basic [50].

The critical pattern: under vegetation occlusion, MVS maintains completeness advantage (75.95% vs. best 3DGS at 65.93%), while 3DGS-MCMC degrades dramatically in accuracy (5.80 mm → 17.36 mm). 3DGS-LumaAI interestingly maintains the best geometric accuracy among 3DGS methods under occlusion (7.58 mm) despite being the worst without occlusion, suggesting its point cloud structure handles occluded regions differently.

5.2 UAV Imagery M3C2 Analysis (MDPI Sensors 2025)

A peer-reviewed comparative study (PMC12115230; MDPI Sensors 25(10):2995) evaluated Nerfacto, Instant-NGP, and Splatfacto on UAV imagery using M3C2 (Multiscale Model-to-Model Cloud Comparison) analysis [28] [44]. SfM-MVS GCP validation for the photogrammetric reference: Study Area 1 RMSE X: 0.026 m, Y: 0.035 m, Z: 0.141 m; Study Area 2 RMSE X: 0.020 m, Y: 0.017 m, Z: 0.240 m [44].

Splatfacto rendering quality (Study Area 2, downscale 8, maximum config): PSNR 28.93, SSIM 0.9180, LPIPS 0.0561—substantially outperforming Nerfacto (PSNR 20.57, SSIM 0.4954) and Instant-NGP (PSNR 24.80, SSIM 0.7715) [44]. M3C2 geometric error: Splatfacto ~0.2 m average in Study Area 2 (best among neural methods), while MVS reference achieved sub-decimeter accuracy. Point clouds from NeRF-based methods become noisier as image resolution decreases; Splatfacto provides more compatible results with the reference point cloud along cross-sections [44].

5.3 Arizona State University Indoor/Outdoor Comparison (arXiv:2508.07483)

A direct comparison using indoor iPhone 13 (1600×1200) and outdoor Canon EOS R (6240×4160) datasets found that Gaussian Splatting demonstrated higher SSIM, PSNR, and lower LPIPS values than photogrammetry in both settings [51] [73]. Spatial resolution: 0.707 lp/mm (3DGS) versus 0.561 lp/mm (photogrammetry) using USAF resolution charts [73]. The performance gap was larger for outdoor scenes.

5.4 Metric Error on Transparent Surfaces (MDPI Sensors 25(14):4410)

This study provides the clearest demonstration of the visual/geometric quality disconnect [74] [114]:

Method	C2M RMSE (m)	Std. Dev. (m)	Completeness	PSNR (dB)
2DGS	0.00121	0.0005 (lowest)	96.43%	34.91
3DGS + SuGaR	0.00125	0.0011	99.62%	36.06
Agisoft Metashape	0.00161 (worst)	0.0008	16.96% (worst)	—

3DGS achieves superior rendering (PSNR 36.06 dB vs. 2DGS's 34.91 dB) yet 2DGS demonstrates better geometric conformity (RMSE 0.00121 m vs. 0.00125 m) and more spatially consistent accuracy (lower standard deviation). Metashape's 16.96% completeness on transparent surfaces versus 96–99% for Gaussian methods represents a fundamental capability gap in specific material classes [74].

5.5 Multi-Dimensional Performance Overview

3DGS vs. Classical Photogrammetry: Multi-Dimensional Comparison

Qualitative comparison based on aggregated findings from ISPRS 2025, GauU-Scene V2, and multiple 2024-2026 studies. Scale 0-100 (high

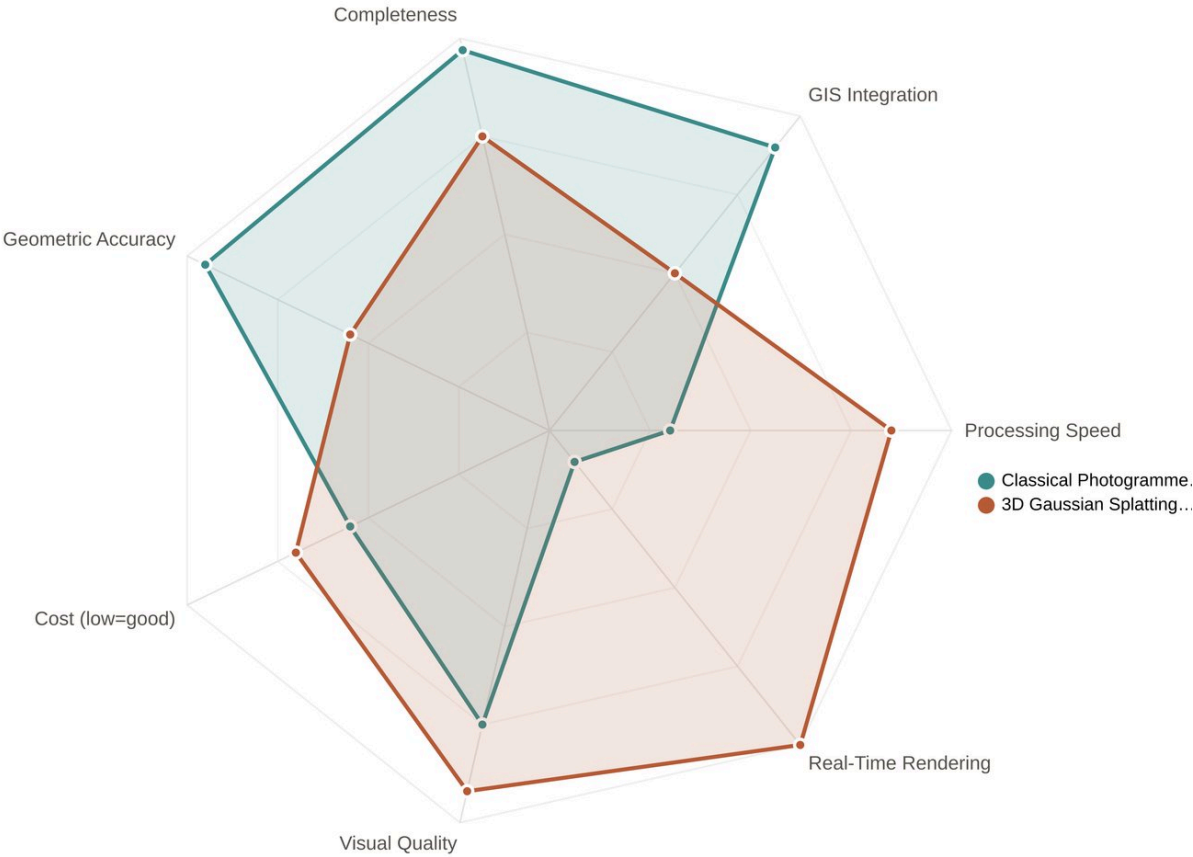


Figure 3: 3DGS vs. Classical Photogrammetry: Multi-Dimensional Comparison: Qualitative comparison based on aggregated findings from ISPRS 2025, GauU-Scene V2, and multiple 2024-2026 studies. Scale 0-100 (higher = better).

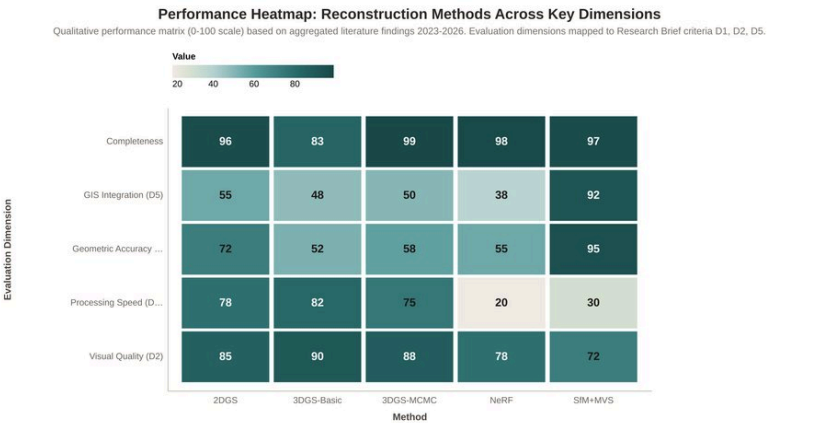


Figure 4: Performance Heatmap: Reconstruction Methods Across Key Dimensions: Qualitative performance matrix (0-100 scale) based on aggregated literature findings 2023-2026. Evaluation dimensions mapped to Research Brief criteria D1, D2, D5.

Summary of directional advantages:

Dimension	Classical SfM/MVS Advantage	3DGS Advantage
Metric geometric accuracy	1.43 mm vs. 4.74 mm RMSE [50]	—
Completeness under occlusion	75.95% vs. 65.93% [50]	—
Rendering PSNR/SSIM/LPIPS	—	+6–9 dB PSNR over NeRF-based [44] [101]
Real-time rendering	—	100–450 FPS vs. offline mesh [43] [96]
Training speed	—	35 min vs. 2–3 hr MVS [40] [46]
Transparent object completeness	—	96–99% vs. 17% [74]
Explicit mesh output (CAD/GIS)	Native triangulation export	Requires SuGaR/2DGS post-processing
GIS integration (2026)	Mature (QGIS, ArcGIS, GeoTIFF)	Emerging (glTF KHR, 3D Tiles, QGIS 4.4 planned) [59] [60] [66]
Regulatory acceptance for surveying	Established	Not yet standardized

6. Georeferencing 3DGS Models: From Research Gap to Available Solutions

Between 2025 and early 2026, the question of georeferencing 3DGS transitioned from a research gap to multiple available solutions, though real-world physical testfield validation remains absent from the literature.

6.1 GeoRefGS: Most Relevant Academic Solution

Hou, J., Zhang, X., Li, H., & Cui, S. (2026). "GeoRefGS: Towards Georeferenced 3D Gaussian Splatting from Unmanned Aerial Vehicle Platforms." *Drones*, 10(3):195. DOI: <https://doi.org/10.3390/drones10030195>. Published March 11, 2026. Special Issue: Advances in Perception, Communications, and Control for Drones: 2nd Edition. Authors: Early Warning Academy Wuhan + Unit 93160 of PLA Beijing [63] [102].

GeoRefGS integrates georeferencing as an intrinsic training constraint rather than post-processing, through: (1) a learnable similarity transformation matrix T connecting local model space to global geographic coordinates; (2) decoupled multi-view geolocation with explicit uncertainty modeling via covariance matrices; (3) multi-tiered geographic loss functions combining absolute positioning loss, GCP constraint loss, and relative geometric constraint loss; and (4) an adaptive weighting mechanism preventing geographic gradients from dominating early optimization.

Accuracy results:

Configuration	Planar Error (m)	Vertical Error (m)	PSNR (dB)
A: 3DGS baseline (no georef.)	Ungeoreferenced	—	32.14
B: Similarity transform T only	25.1 m	—	22.65
C: + Absolute positioning loss	0.123 m	—	31.12
D: + GCP constraint loss	0.0145 m	—	34.92
E: Full GeoRefGS (adaptive weighting)	0.054 m	0.053 m	39.91
SfM-BBA + 3DGS (traditional)	0.32 m	—	—

The GCP constraint loss alone reduces planar error from 25.1 m to 0.0145 m—demonstrating that GCPs are essential for achieving sub-decimeter accuracy in any 3DGS georeferencing workflow [63]. The full method's 0.054 m planar accuracy represents a reduction from the 0.32 m SfM-BBA baseline (83% improvement).

Critical limitation: GeoRefGS validation was conducted in a simulated Unity environment, not real drone captures [63]. This is the methodological gap the proposed thesis would directly address.

6.2 Commercial and Open-Source Solutions

Pix4D (PIX4Dcatch): Announced georeferenced 3DGS integration on October 7, 2025 [64]. Supports 50–2,000 image datasets with or without RTK. Unlike traditional photogrammetry, 3DGS preserves visibility of thin structures (scaffolding, reinforcement meshes) that disappear in mesh noise [64] [65].

GPS-EXIF Open-Source Pipeline (manudelu/georeferenced_gsplat): Extracts GPS metadata from drone images, aligns via COLMAP SfM to ENU coordinates using the first image as reference. Compatible with Cesium Ion, CloudCompare, and Unreal Engine [62].

Cesium Ion End-to-End: Automatic reconstruction as Mesh, PointCloud, or 3DGS. Microsoft Redmond Campus example: 20,169 photos (427.7 Gigapixels), 3.7 km², 110 million splats, 3 cm GSD [70].

Smart UAS-GCP concept: Combines PPK, RTK, and classical GCP in a single transportable unit with centimeter-level accuracy for all three correction techniques [89].

6.3 GIS Integration Status (2026)

Platform	Status	Capability
QGIS 3.34+	Production	3D Tiles via QgsTiledSceneLayer (North Road/Lutra Consulting, Cesium Ecosystem Grant) [59]
QGIS 4.4 (planned)	In development	Native 3DGS extension sponsored project [60]
ArcGIS Pro	In development	3D Tiles (WGS84 only), I3S via CesiumJS; EGM96 vertical default [61]
CesiumJS	Production	Full LOD 3DGS streaming (April 2026 update) [70]
CloudCompare	Production (indirect)	Via PLY import for M3C2/geometric analysis
CityGML	Not documented	No integration found in any source
Potree	Not documented	No integration found in any source

The glTF KHR_gaussian_splatting Release Candidate (February 2026) defines storage of XYZ position, quaternion rotation (encoded as 10-bit signed integers for the three smallest components), anisotropic ellipsoid scale, opacity, and spherical harmonics [66] [67] [69]. The SPZ compression format reduces file size up to 90% versus uncompressed PLY while maintaining visual quality [66] [68].

7. Physical Test Field Design: GCP Specifications for 3DGS Benchmarking

Establishing a physical GCP testfield measured by total station or RTK GNSS is the methodological core of the proposed thesis and the primary differentiator from existing simulation-based studies.

7.1 GCP Target Design

GCP markers should employ a checkerboard or L-shaped pattern (not X-pattern—X-patterns can introduce up to 10 cm of measurement uncertainty [77]) with matte surfaces to avoid glare and reflections [76] [81]. The critical sizing relationship: **target size must be 5–10× the Ground Sampling Distance** [77] [78] [79]. Concrete examples: at 2 cm/pixel GSD, targets should be 10–20 cm wide; at 5 cm/pixel GSD, minimum 25–50 cm. The absolute minimum for typical UAV heights is 0.5 m × 0.5 m [78]. For permanent painted markings, minimum 600 mm per direction [77].

7.2 GCP Density and Distribution

Survey Area	Minimum GCPs	Recommended	Notes
<5 ha (12 acres)	3	5–8	Baseline triangulation
5–20 ha	5	7–10	Standard projects
20–80 ha	8	12–15	Larger coverage
>80 ha	10	15–20+	Complex geometry

Distribution principles: perimeter placement first, then interior, with GCPs at highest and lowest elevation points for significant terrain relief [75] [77] [79]. Minimum 100 m spacing between GCPs [75] [76]. For linear corridor projects (2.1 km × 190 m road study [83]): optimal configuration uses alternating offset pattern on both sides plus endpoint pairs (9–11 GCPs, 4.3–5.2 GCPs/km) achieving XY RMSE 0.028–0.029 m and Z RMSE 0.055–0.057 m. General density: 0.5–1.8 GCPs/hectare depending on terrain [83].

More than 5–10 GCPs shows diminishing returns—beyond this count, accuracy improvements follow a logarithmic curve [78] [82]. Quality Control Points (QCPs), deliberately excluded from processing and used for independent validation, are methodologically essential—without them, reported accuracy overestimates by 3–6× [31] [77] [82].

7.3 Measurement Standards

RTK GNSS equipment requirement: Horizontal and vertical accuracy ≤ 0.02 m (2 cm); Fix solution with estimated error <3 cm; two separate sessions of 30 measurements at 1-second intervals [94]. RTK systems like Leica GS-Series, Topcon HiPer-Series, Emlid Reach RS2, Stonex S-Series, and Trimble R-Series achieve 1–3 cm horizontal in normal conditions [93].

PPK (Post-Processing Kinematic): Equivalent centimeter accuracy to RTK, applied post-fieldwork; preferred in areas with poor cellular connectivity; supports baselengths up to 100 km [92] [94].

Total station: Millimeter-level angular precision; recommended for sub-centimeter GCP accuracy requirements or GPS-challenged locations [92] [93].

Pole setup: Antenna height precisely measured and entered into rover; surveying rod absolutely straight; level bubble centered during data capture; rover tip precisely centered on target crosshair [93] [94]. GCP center must localize to pixel-level precision in drone images, with minimum 5–6 image overlaps and at least 150-pixel distance from image boundaries [94].

With RTK/PPK-equipped drones: Only 4–5 GCPs per square mile (2.6 km²) required for elevation variation control; high-contrast urban environments may need only 3 well-distributed control points [81].

7.4 Xianning Test Field Reference Case

A large test field with 118 GCPs established via static GPS surveys achieved approximately 10 cm accuracy across 120 km × 100 km, with 4 corner GCPs proving optimal for satellite imagery control—additional points provided minimal improvement [80]. This provides a useful reference for satellite imagery validation but is at a completely different scale from UAV testfield design.

8. Benchmark Datasets for External Validation

8.1 Comparison of Standard Benchmarks

Dataset	Ground Truth	Scale	Primary Use	Aerial/UAV Suitability
Tanks & Temples (SIGGRAPH 2017) [86] [87]	FARO laser (0.1 mm @ 10.2 m)	14 scenes, object-to- building scale	F-score (precision/recall)	Indoor/outdoor, no UAV- specific
ETH3D (CVPR 2017) [84] [85]	Laser scan	13+12 scenes, 24 MP DSLR	Accuracy/Completeness/F1	Indoor+outdoor, diverse
DTU [90] [91]	Structured light	124 scenes, 1600×1200	Accuracy/Completeness/Overall	Small objects, lab- controlled
MipNeRF360 [3]	None (rendering only)	9 scenes (5 outdoor+4 indoor)	PSNR/SSIM/LPIPS	No geometric GT— rendering benchmark only
BlendedMVS (CVPR 2020) [98] [99]	Mesh rendering	113 scenes, 17,818 images	Training generalization	Urban/architecture variety
GauU-Scene V2 [101]	LiDAR (627.5M points)	6 scenes, 6.668 km ²	Chamfer distance + PSNR	Most relevant for UAV 3DGS
NERFBK (ISPRS 2023) [88] [100]	Various	Heritage, transparent, aerial	Multi-category comparison	Includes aerial

The Tanks and Temples benchmark (Knapitsch, Park, Zhou, Koltun, Intel Labs) acquires ground truth via FARO industrial scanner at 976,000 points/second with registration uncertainty of 0.1 mm at 10.2 m and 0.3 mm at 22.7 m [87]. Evaluation uses F-score (harmonic mean of precision and recall) at distance threshold τ , with voxel resampling at $\tau/2$ [87].

ETH3D evaluation uses voxel-based protocol to prevent manipulation (uniform space-filling would artificially inflate completeness), computing accuracy and completeness per voxel individually [85]. Distance thresholds from 1 cm to 50 cm [85].

For the proposed thesis, **GauU-Scene V2** is the most directly relevant external dataset, providing both rendering quality and geometric accuracy (Chamfer distance against LiDAR) for UAV-scale scenes with drone-captured imagery [101].

8.2 Processing Time and Hardware Context

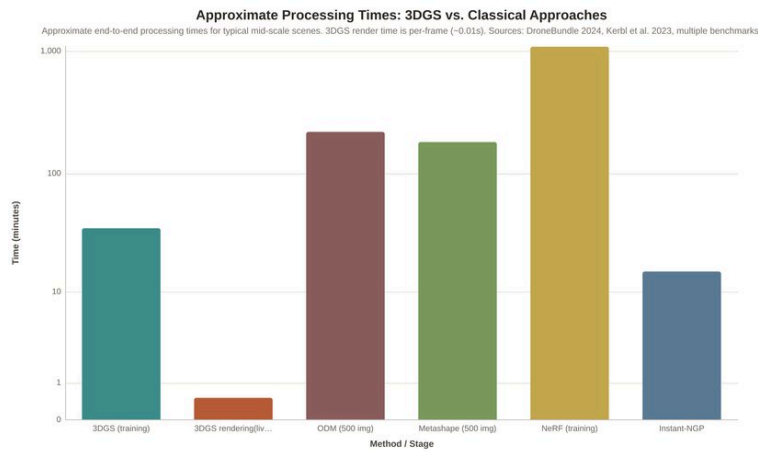


Figure 5: Approximate Processing Times: 3DGS vs. Classical Approaches: Approximate end-to-end processing times for typical mid-scale scenes. 3DGS render time is per-frame (~0.01s). Sources: DroneBundle 2024, Kerbl et al. 2023, multiple benchmarks.

Classical photogrammetry hardware for Agisoft Metashape: minimum 4-12 cores @ 2.0+ GHz, 16-32 GB RAM, NVIDIA/AMD GPU with 1024+ unified shaders; enhanced configuration 32-128 GB RAM, RTX 3080 or equivalent [71]. For 500-image building facade: COLMAP + OpenMVS ~6 hours (48 GB RAM), Metashape ~3 hours (64 GB RAM) [72].

3DGS VRAM requirements from experimental data [8]:

Dataset Type	Method	VRAM (GB)
Medium outdoor building	Proprietary optimized	0.28
Medium outdoor building	GaussianPro	0.52
Complex sparse-view	Proprietary optimized	2.98
Complex sparse-view	GaussianPro	4.39

Faster-GS achieves up to 5× training acceleration in optimal scenarios, with typical improvements of ~30% on Mip-Splatting benchmarks (reducing from ~16 minutes to ~11 minutes) and VRAM reduction of ~30% (8.0 GiB → 6.1 GiB) [95] [97]. FlashGS (CVPR 2025) achieves 450.5 FPS at 1080p for the Truck scene and 289.0 FPS at 4K [96].

9. Confirmed Research Gaps and Thesis USP

The literature review confirms several research gaps relevant to the proposed PA/thesis work, with one gap directly actionable as the thesis's unique scientific value:

Confirmed Gap 1 (Thesis USP): No published study has combined a physical GCP testfield measured by total station or RTK GNSS with 3DGS reconstruction and evaluated metric accuracy against surveying-grade ground truth. GeoRefGS (the closest published study) used a Unity-simulated environment [63]. This gap is directly addressable at THWS Würzburg with campus survey equipment and a drone dataset over a designed testfield.

Confirmed Gap 2: No standardized cost-speed-accuracy benchmark integrating all three dimensions simultaneously for UAV workflows comparing 3DGS vs. classical on identical datasets exists in the peer-reviewed literature. Software pricing analyses (DroneBundle [46]) and processing time benchmarks [40] [72] exist separately but are never synthesized against geodetic accuracy on the same data.

Confirmed Gap 3: Skyfall-GS has not been applied to oblique UAV drone footage. Applying or adapting the Skyfall-GS satellite-imagery architecture to drone data would be technically novel, but risk assessment is required—the pseudo-camera depth supervision component is specifically designed for limited-parallax satellite views and may not transfer without significant modification [5].

Confirmed Gap 4: D3/D4 evaluation dimensions—manual object identifiability and ML-based object detection on 3DGS renders versus photogrammetric models—have no systematic treatment in the literature. The 3DGS-QA dataset (225 degraded reconstructions, 15 object types [52]) addresses perceptual quality but not object detection performance. This constitutes an effectively blank slate for novel evaluation contribution.

Confirmed Gap 5: The inverse relationship between image metrics and geometric accuracy (GauU-Scene finding) has not been made the central research question in a geodetically motivated paper. Designing a study that explicitly tests this relationship with GCP-derived ground truth would constitute an original methodological contribution.

Notable non-gap (addressed): Georeferencing methodology for 3DGS is no longer primarily a research gap—GeoRefGS, Pix4D commercial release, GPS-EXIF pipeline, and glTF standards collectively address the integration question. The remaining gap is specifically in real-world field validation, not in the methods themselves [63] [64] [69].

10. Satellite Gaussian Splatting Context: EOGS and Skyfall-GS Positioning

For completeness in understanding the satellite-specific literature, two additional papers provide context:

Aira, L.S., Facciolo, G., & Ehret, T. (2025). "Gaussian Splatting for Efficient Satellite Image Photogrammetry." *CVPR 2025*, pp. 5959–5969. arXiv:2412.13047 [107] [108] [109]. EOGS (Earth-observation Gaussian Splatting) achieves approximately 300× speedup versus EO-NeRF (reconstruction in ~4.60 minutes versus 5.76–7.39 hours) with comparable or superior MAE values [108] [6]. Key technical innovations: local pushbroom approximation (mean error ≈ 0.012 pixels), physically-based shadow mapping, and three regularization terms (sparsity with $\alpha_{min} = 0.0025$ achieving 2× speedup, view-consistency, and opacity regularization) [6]. Elevation MAE on test scenes: JAX004 3.31 m, JAX068 6.67 m, JAX260 6.41 m, OMA212 9.08 m, OMA315 6.38 m, IARPA001 5.90 m, IARPA002 13.79 m, IARPA003 14.83 m [6]. Performance is best in high-coverage regions (>80% image overlap: MAE 0.8–1.0 m) and degrades under dense vegetation [6].

Skyfall-GS's FIDCLIP performance advantage over EOGS (FIDCLIP 27.35 vs. 87.74 on DFC2019) reflects the diffusion-based generative refinement stage adding photorealistic detail not achievable through pure reconstruction [5]—confirming that the comparison is between a reconstruction system (EOGS) and a generation system (Skyfall-GS), not equivalent methods on the same task.

11. Methodological Recommendations for the Proposed PA/Thesis

11.1 Recommended Pipeline Stack

Classical photogrammetry (free/available):

- SfM: COLMAP (CLI, free, THWS-compatible)
- MVS: OpenMVS or ODM (WebODM, free)
- Georeferencing: GPS-EXIF pipeline [62] or Metashape Pro (THWS perpetual license available)
- QA/geometric analysis: CloudCompare (M3C2, Hausdorff, C2M distance)

3DGS:

- gsplat library (PyTorch+CUDA, 4× less memory than INRIA reference [16])
- Nerfstudio Splatfacto for comparable NeRF baseline
- DJI Terra V5.0 for commercial 3DGS baseline (if Matrice 4E available)
- GeoRefGS framework for georeferencing integration

Optional:

- DroneSplat for dynamic scene handling [106]
- SuGaR for mesh extraction from 3DGS [103]
- 2DGS for surface-accurate geometric reconstruction [53]

11.2 Recommended Evaluation Metrics

D1 (Geometric accuracy): RMSE_X, RMSE_Y, RMSE_Z separately against GCP ground truth; M3C2 distance (CloudCompare); Chamfer distance against LiDAR (if available); C2M distance for object-scale evaluation

D2 (Visual quality): PSNR, SSIM, LPIPS—all three are community standard. LPIPS provides perceptual similarity via learned features; always report all three because GauU-Scene demonstrates their potential anti-correlation with geometric accuracy [101]

D3 (Manual object identifiability): Structured annotation protocol over rendered views at defined altitude increments; inter-rater reliability assessment

D4 (ML object detection): Run a standard detector (e.g., YOLOv8) on rendered views from both methods at matched resolutions; report mAP at IoU thresholds 0.5 and 0.75; this constitutes a novel contribution

D5 (Practicality): End-to-end wall-clock time; peak VRAM consumption; model size in MB; software licensing cost (€/year); hardware cost amortized per project

***Critical methodological note:** Always report both rendering metrics (PSNR/SSIM/LPIPS) and geometric metrics (RMSE, Chamfer, M3C2) simultaneously. The GauU-Scene finding of inverse correlation between these two families of metrics [101] means that reporting only one category will mislead readers about the method's actual fitness for geodetic applications.*

11.3 Test Field Design for Campus Deployment

For a THWS campus testfield (~0.5–2 ha typical):

- **Recommended GCP count:** 8–12 GCPs (5 perimeter + 3–7 interior) + 4–6 independent QCPs
- **Target dimensions:** 50 cm × 50 cm minimum; for GSD 2 cm/pixel, 20 cm patterns; use L-shape checkerboard with matte finish [77]
- **Measurement:** RTK GNSS (Fix solution, HRMS <0.02 m, two sessions × 30 observations at 1-second intervals) [94]
- **Distribution:** Corners and center for rectangular areas; GCPs at highest/lowest elevation; minimum 100 m between points; QCPs not used in processing [79] [82]
- **Flight parameters:** 75–85% front overlap, 60–70% side overlap; include 10–12% oblique imagery at 20° inclination for improved vertical accuracy [29]; constant shutter 1/800 s, ISO 200, f/4.5 [31]

11.4 Thesis Framing Clarification

The research brief states Skyfall-GS is "optimized for oblique aerial and satellite images." The actual published paper evaluates exclusively nadir satellite data (WorldView-3, GoogleEarth) and has not been tested on UAV footage [5]. The thesis should either: (a) reframe the comparison to use a drone-specific 3DGS pipeline (DroneSplat, standard 3DGS, DJI Terra V5.0) with Skyfall-GS mentioned as satellite-specific context; or (b) design an original experiment adapting Skyfall-GS to drone data, explicitly noting this constitutes novel application research beyond the paper's scope. Option (a) is lower risk for a bachelor's thesis timeline; option (b) would constitute the strongest original scientific contribution.

12. Software Cost Summary

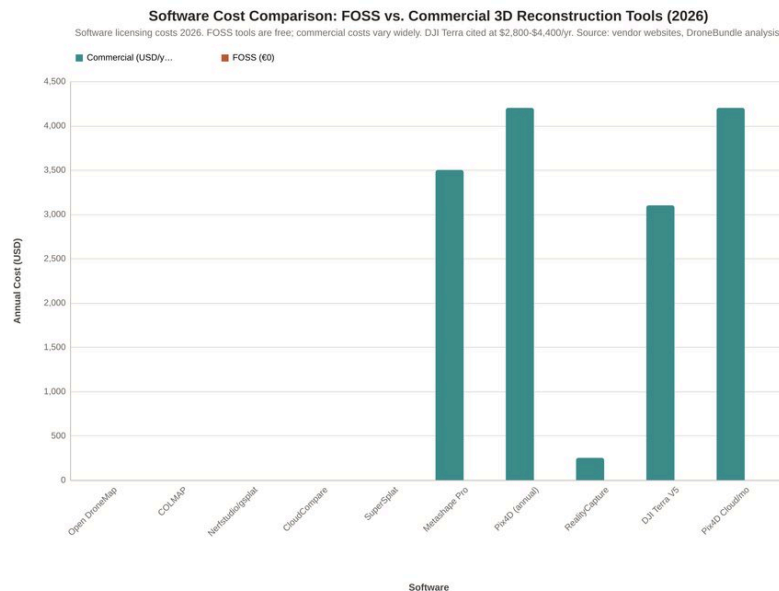


Figure 6: Software Cost Comparison: FOSS vs. Commercial 3D Reconstruction Tools (2026): Software licensing costs 2026. FOSS tools are free; commercial costs vary widely. DJI Terra cited at \$2,800-\$4,400/yr. Source: vendor websites, DroneBundle analysis.

Software	Cost (2026)	Category	Best Use Case
COLMAP	\$0	FOSS	Research/maximum precision
OpenDroneMap	\$0	FOSS	UAV-specific, web publication
Nerfstudio/gsplat	\$0	FOSS	3DGS/NeRF research
CloudCompare	\$0	FOSS	Geometric analysis/M3C2
Agisoft Metashape Pro	\$3,499 perpetual	Commercial	Survey/engineering accuracy
Pix4D Mapper	\$4,200/year	Commercial	GCP workflows, GIS integration
DJI Terra V5.0	\$2,800-\$4,400/year	Commercial	Commercial drone workflows
RealityCapture	\$0.25-\$1.00/image (PPI)	Commercial	Mesh/VFX, speed priority
DroneDeploy	\$4,000-\$12,000/year	Cloud	High-volume commercial

For a thesis comparison, the recommended cost-neutral stack is COLMAP + ODM + gsplat (all \$0) with Metashape Pro as the commercial baseline under THWS institutional license. Total additional hardware cost beyond standard laptop: approximately \$0 if THWS GPU workstations are available; 3DGS training requires minimum 6–8 GB VRAM for medium-scale scenes, 24 GB for large aerial datasets [8] [97].

Conclusion: Strategic Research Positioning

The proposed PA/thesis occupies a validated and genuinely novel research position. The literature through mid-2026 demonstrates that: (1) 3DGS consistently outperforms classical photogrammetry on visual quality metrics but consistently underperforms on geometric accuracy in peer-reviewed geodetic tests [50] [101] [74]; (2) georeferencing solutions for 3DGS exist but have only been validated in simulation [63]; (3) the relationship between rendering metrics and geometric fidelity is systematically inverse for 3DGS under UAV conditions [101]—a finding not yet tested against GCP-derived surveying ground truth in a controlled physical testfield.

The thesis's distinctive contribution lies precisely at the intersection of these three observations: a physical GCP testfield provides the ground truth geometry that simulation-based studies lack, the simultaneous PSNR/SSIM/LPIPS and RMSE/M3C2/Chamfer evaluation explicitly tests the metric-geometry disconnect on drone data, and the cost-speed-accuracy synthesis addresses a practical gap in the applied geomatics literature. The addition of ML object detectability as D4 would be a further genuinely original element with no direct precedent in any reviewed study.

Skyfall-GS's satellite-specific architecture should be acknowledged in the introduction and related work as an important advance in satellite urban reconstruction, positioned as context for understanding 3DGS adaptation to aerial imaging challenges—rather than as the primary pipeline under evaluation for UAV drone scenarios.

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